

Practical Task 1: Supervised Learning

Ordinary least squares on the California Housing dataset

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1. Task

Reproduce the supervised-learning example from the lecture on a dataset other than Boston Housing. The lecture figure is a scatter of one numeric input against a continuous target, with an ordinary least squares (OLS) line drawn through the points. This report uses the same setup.

2. Dataset

The table is California Housing (Pace and Barry, 1997): 20,640 census block groups from the 1990 US Census. sklearn documents it as the replacement for Boston Housing. I downloaded the StatLib archive from Figshare (<https://ndownloader.figshare.com/files/5976036>) and stored the prepared table as `data/california_housing.csv`.

Boston Housing predicted median house value from average rooms. On this table the rooms analogue, AveRooms, correlates 0.15 with price. Median income (MedInc) correlates 0.69, so that is the feature used for the lecture-style plot and the one-feature model. The eight-feature model is the same estimator with the remaining columns included.

There are no missing values. MedHouseVal is stored in units of \$100,000 and is capped at 5.0 (\$500,000) for 992 block groups (4.8% of the table). Boston Housing had the same kind of cap at \$50,000. Extreme values of AveRooms (max 142) and AveOccup (max 1,243) were left in the table.

3. Features (X) and target (y)

Task type: **regression**. The target is continuous.

Variable	Role	Meaning
MedInc	input	median income in the block group, tens of thousands of USD
HouseAge	input	median house age, years
AveRooms	input	average rooms per household

Variable	Role	Meaning
AveBedrms	input	average bedrooms per household
Population	input	block-group population
AveOccup	input	average household size
Latitude, Longitude	input	block-group location
MedHouseVal	target (y)	median house value, units of \$100,000

4. Relationship between input and target

Figure 1, panel A, is the Boston-style view: one numeric input against the continuous target. Panel B is average rooms, the feature that usually played that role in the lecture. Each panel is a random sample of 4,000 points so the cloud is readable. The fitted models use all 20,640 rows.

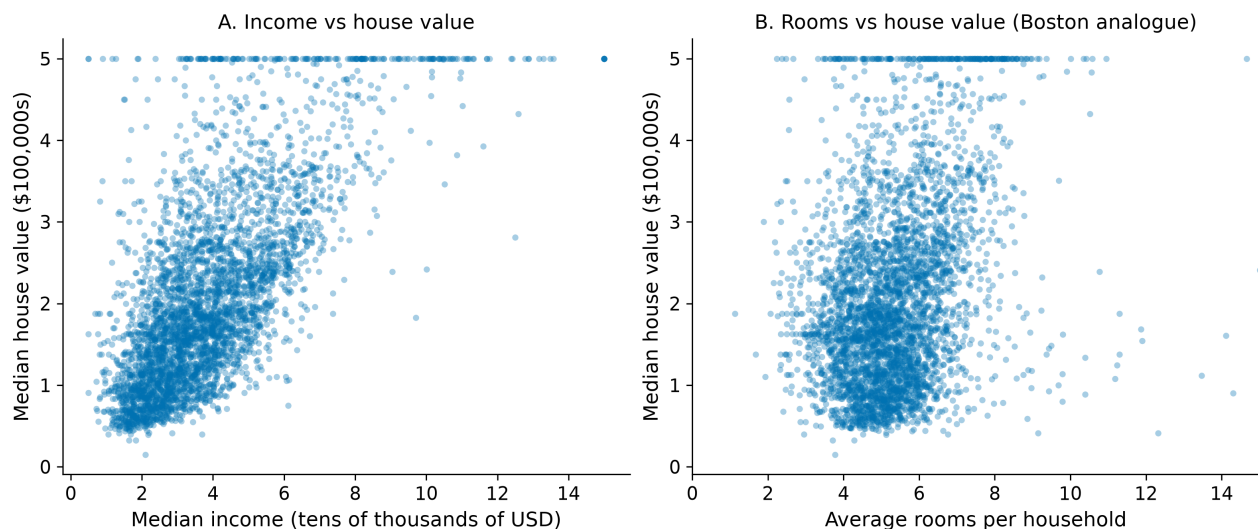


Figure 1: Median house value against median income (A) and against average rooms (B). Sample $n = 4,000$ of $N = 20,640$ block groups. House value is in units of \$100,000; income is in tens of thousands of USD.

Panel A rises with income, then hits the \$500,000 ceiling. The cloud widens as income grows: the same income is compatible with a wide range of prices. Panel B, cut to a readable rooms range, is almost a vertical smear. A rooms-only line is a poor one-feature model on this table.

5. Model

The estimator is ordinary least squares (`sklearn.linear_model.LinearRegression`), the same class of model as the lecture. 20% of block groups were held out ($n_{\text{test}} = 4,128$). The split is random with `random_state=42`. The model was fit only on the remaining 80% ($n_{\text{train}} = 16,512$).

The one-feature model uses `MedInc` alone. The eight-feature model is still OLS. Feature scaling sits inside a Pipeline and is fit on the training split only, so population and latitude are not treated as if they shared a scale.

The fitted one-feature equation is

$$\text{MedHouseVal} = 0.445 + 0.419 \times \text{MedInc}.$$

Each extra \$10,000 of block-group median income is associated with \$41,934 higher median house value.

6. Predictions

Predictions are the OLS fitted values on the held-out test block groups. Figure 2 is the lecture figure: observed y as points, predicted y as the line. The line was fit on the training split, not on these points.

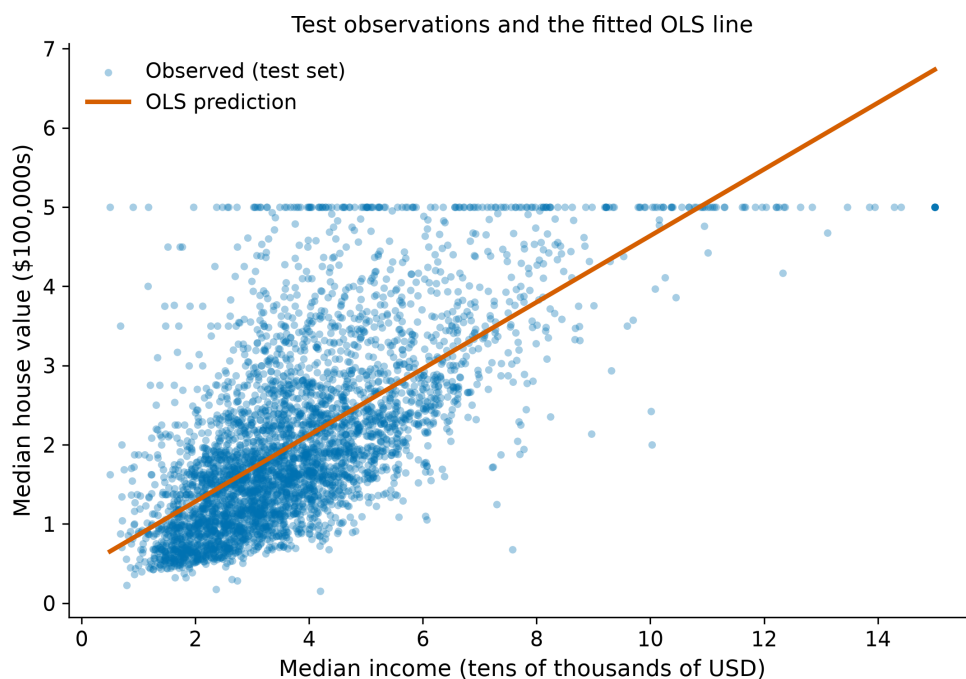


Figure 2: Test observations ($n = 4,128$) and the OLS line $\text{MedHouseVal} = 0.445 + 0.419 \times \text{MedInc}$. The horizontal stripe at 5.0 is the \$500,000 cap. The line has no upper bound and continues above that cap.

7. Evaluation

Regression is scored with RMSE (typical residual size, in the target's units), MAE (same, less pulled by large misses), and R^2 (fraction of variance matched on the test set). Accuracy is a classification metric and does not apply.

Model	n_{test}	RMSE	RMSE (USD)	MAE (USD)	R^2
OLS, AveRooms only	4,128	1.137	113,681	88,915	0.014
OLS, MedInc only	4,128	0.842	84,209	62,991	0.459
OLS, eight features	4,128	0.746	74,558	53,320	0.576

The mean test house value is \$205,500. For the income-only model, MAE is about 31% of that mean.

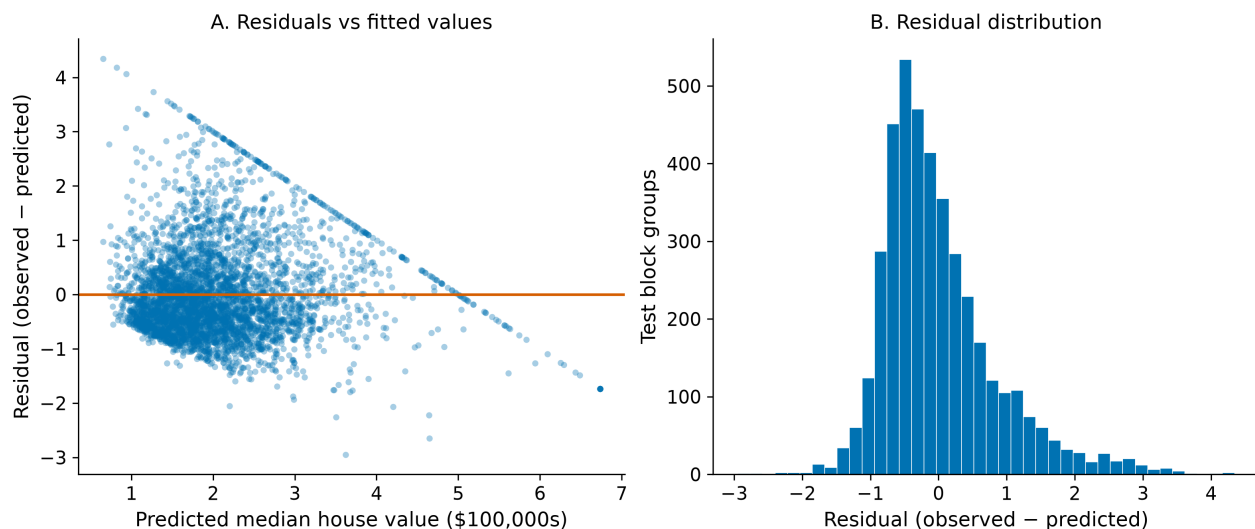


Figure 3: Test residuals for the income-only model. Panel A: residual against fitted value. The downward diagonal is the \$500,000 cap (observed = 5.0). Panel B: residual histogram. Residual variance grows with the fitted value, so the OLS constant-variance assumption does not hold.

8. Conclusion

The one-feature model learned a positive linear association between block-group median income and median house value: $R^2 = 0.459$ on 4,128 held-out block groups, RMSE about \$84,200. That is a usable average trend, not a tight fit. The scatter around the line is wide, it widens at higher incomes, and 4.8% of rows sit at a \$500,000 cap that OLS cannot represent, so the fitted line keeps rising past \$500,000.

AveRooms, the usual Boston Housing x -variable, gives test $R^2 = 0.014$ here. Rooms are not the main price signal in 1990 California census data. Adding the other seven features raises test R^2 to 0.576 and cuts RMSE to about \$74,600. After scaling, location and income carry most of the weight. AveRooms turning negative is collinearity with bedrooms and income, not a claim that extra rooms lower price.

The coefficients describe a linear association in 1990 census geography. They are not an estimate of what would happen if income changed. Figure 2 is the lecture plot: a positive slope through a noisy cloud. Income is the feature that line actually tracks.

Reference

Pace, R. Kelley, and Ronald Barry. 1997. "Sparse Spatial Autoregressions." *Statistics & Probability Letters* 33: 291–297.

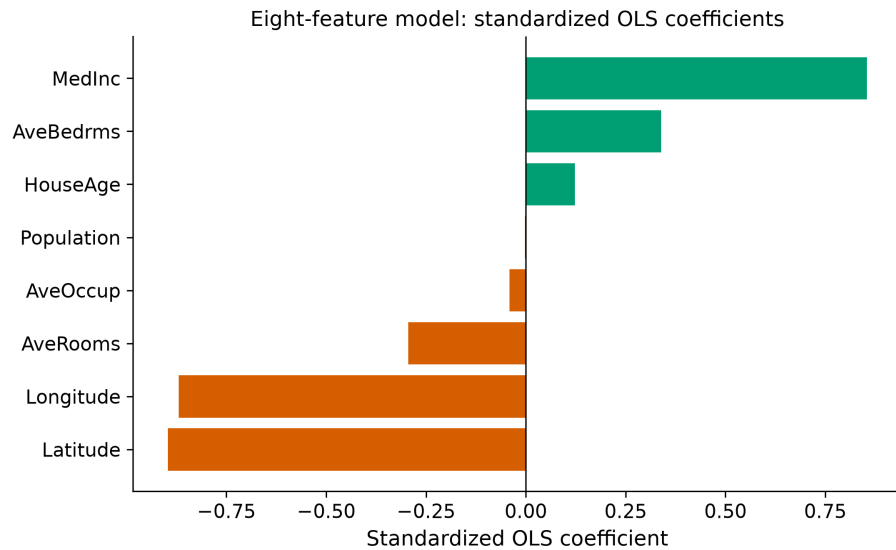


Figure 4: Standardized coefficients of the eight-feature OLS model (features scaled on the training split only). Latitude, longitude, and income dominate. AveRooms is negative once bedrooms and income are in the model.

Appendix. Code

The runnable notebook with outputs is `Report.ipynb` in this folder. The analysis that produced the numbers and figures above is:

```
from pathlib import Path
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score, root_mean_squared_error
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

df = pd.read_csv("data/california_housing.csv")
y = df["MedHouseVal"]
feature_cols = [
    "MedInc", "HouseAge", "AveRooms", "AveBedrms",
    "Population", "AveOccup", "Latitude", "Longitude",
]

X_train, X_test, y_train, y_test = train_test_split(
    df[feature_cols], y, test_size=0.2, random_state=42
)
model = LinearRegression().fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
print(model.intercept_, model.coef_[0])
print(root_mean_squared_error(y_test, y_pred), r2_score(y_test, y_pred))

full = Pipeline([("scaler", StandardScaler()), ("ols", LinearRegression())])
Xf_train, Xf_test, yf_train, yf_test = train_test_split(
    df[feature_cols], y, test_size=0.2, random_state=42
)
full.fit(Xf_train, yf_train)
print(root_mean_squared_error(yf_test, full.predict(Xf_test)),
      r2_score(yf_test, full.predict(Xf_test)))
```